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Final Exam Practicum Algorithm & Data Structure

Vehicle Class

```
1  package com.finalExam;
2
3  public class Vehicle
4  {
5      String  tnkb,
6             name,
7             type;
8      int     cc,
9             year,
10            dueMonth;
11
12     public Vehicle() {}
13     public Vehicle (String tnkb, String name, String type, int cc, int year, int dueMonth)
14     {
15         this.tnkb      = tnkb;
16         this.name       = name;
17         this.type       = type;
18         this.cc         = cc;
19         this.year       = year;
20         this.dueMonth   = dueMonth;
21     }
22 }
```

Tax Class

```
1  package com.finalExam;
2
3  public class Tax
4  {
5      Vehicle dataVehicle;
6
7      int code,
8          nominal,
9          fine,
10         monthPay;
11
12     public Tax() {}
13     public Tax (Vehicle dataVehicle, int monthPay)
14     {
15         this.dataVehicle    = dataVehicle;
16         this.monthPay       = monthPay;
17         this.nominal        = nominal();
18         this.fine           = fine();
19     }
20
21     int nominal()
22     {
23         if (dataVehicle.type.equalsIgnoreCase("2 wheels"))
24         {
25             if (dataVehicle.cc < 100)
26                 return 100000;
27             else if (dataVehicle.cc > 250)
28                 return 500000;
29             else
30                 return 250000;
31         }
32         else
33         {
34             if (dataVehicle.cc < 1000)
35                 return 750000;
36             else if (dataVehicle.cc > 2500)
37                 return 1500000;
38             else
39                 return 1000000;
40         }
41     }
42 }
```

```

43     int fine()
44     {
45         if (monthPay <= dataVehicle.dueMonth)
46             return 0;
47         else
48         {
49             int fineMonth = monthPay - dataVehicle.dueMonth;
50
51             if (fineMonth > 3)
52                 return 50000 * fineMonth;
53             else
54                 return 50000;
55         }
56     }
57
58     void print()
59     {
60         int totNominal = nominal + fine;
61
62         System.out.printf
63         (
64             format: "%-6s | %-10s | %-12s | %-8s | %-5d | %-7d | %-7d | Rp%-9d | Rp%-7d | Rp%-9d\n",
65             code, dataVehicle.tnkb, dataVehicle.name, dataVehicle.type,
66             dataVehicle.cc, dataVehicle.dueMonth, monthPay, nominal, fine, totNominal
67         );
68     }
69 }

```

Node Vehicle Class

```

1  package com.finalExam;
2
3  public class NodeVehicle
4  {
5      Vehicle dataVehicle;
6      NodeVehicle prev, next;
7
8      public NodeVehicle() {}
9      public NodeVehicle (Vehicle dataVehicle)
10     {
11         this.dataVehicle = dataVehicle;
12         prev = null;
13         next = null;
14     }
15
16     public NodeVehicle(NodeVehicle prev, Vehicle dataVehicle, NodeVehicle next)
17     {
18         this.dataVehicle = dataVehicle;
19         this.prev = prev;
20         this.next = next;
21     }
22 }

```

Node Tax Class

```
1  package com.finalExam;
2
3  public class NodeTax
4  {
5      Tax dataTax;
6      NodeTax prev, next;
7
8      public NodeTax() {}
9      public NodeTax (Tax dataTax)
10     {
11         this.dataTax    = dataTax;
12         prev            = null;
13         next            = null;
14     }
15
16     public NodeTax (NodeTax prev, Tax dataTax, NodeTax next)
17     {
18         this.dataTax    = dataTax;
19         this.prev       = prev;
20         this.next       = next;
21     }
22 }
```

Double LinkedList Vehicle Class

```
1  package com.finalExam;
2
3  public class DLLVehicle
4  {
5      NodeVehicle head, tail;
6
7      public DLLVehicle()
8      {
9          head    = null;
10         tail    = null;
11     }
12
13     boolean isEmpty()
14     {
15         return head == null;
16     }
17 }
```

```
18 void addVehicleData (Vehicle data)
19 {
20     NodeVehicle newNode = new NodeVehicle (data);
21
22     if (isEmpty())
23         head = tail = newNode;
24     else
25     {
26         tail.next      = newNode;
27         newNode.prev    = tail;
28         tail            = newNode;
29     }
30 }
31
32 Vehicle searchVehicleTNKB (String tnkb)
33 {
34     NodeVehicle temp = head;
35
36     while (temp != null)
37     {
38         if (temp.dataVehicle.tnkb.equalsIgnoreCase(tnkb))
39             return temp.dataVehicle;
40         temp = temp.next;
41     }
42
43     return null;
44 }
45
46 }
```

Double LinkedList Tax Class

```
1  package com.finalExam;
2
3  public class DLLTax
4  {
5      NodeTax head, tail;
6
7      public DLLTax()
8      {
9          head    = null;
10         tail    = null;
11     }
12
13     boolean isEmpty()
14     {
15         return head == null;
16     }
17
18     void addTaxTransaction (Tax data)
19     {
20         NodeTax newNode = new NodeTax (data);
21
22         if (isEmpty())
23         {
24             data.code = 1;
25             head = tail = newNode;
26         }
27         else
28         {
29             data.code = tail.dataTax.code + 1;
30             tail.next    = newNode;
31             newNode.prev  = tail;
32             tail         = newNode;
33         }
34     }
35 }
```

```

36 void displayAll()
37 {
38     if (isEmpty())
39         System.out.println(x: "Tax Transaction is empty!!");
40     else
41     {
42         InsertionSortAscending();
43
44         System.out.printf
45         (
46             format: "%-6s | %-10s | %-12s | %-8s | %-5s | %-7s | %-7s | %-11s | %-9s | %-11s\n",
47             ...args: "Code", "TNKB", "Name", "Type", "CC", "Due Mon", "Pay Mon", "Nominal", "Fine", "Total Pay"
48         );
49
50         NodeTax temp = head;
51         int totalTransaction = 0;
52
53         while (temp != null)
54         {
55             temp.dataTax.print();
56             totalTransaction += (temp.dataTax.nominal + temp.dataTax.fine);
57             temp = temp.next;
58         }
59         System.out.println("Total Transaction: Rp " + totalTransaction);
60     }
61 }
62

```

```

63 void InsertionSortAscending()
64 {
65     NodeTax current = head.next;
66
67     while (current != null)
68     {
69         Tax letter = current.dataTax;
70         String letterName = letter.dataVehicle.name;
71
72         NodeTax j = current.prev;
73
74         while (j != null && j.dataTax.dataVehicle.name.compareToIgnoreCase(letterName) > 0)
75         {
76             j.next.dataTax = j.dataTax;
77             j = j.prev;
78         }
79
80         if (j == null)
81             head.dataTax = letter;
82         else
83             j.next.dataTax = letter;
84
85         current = current.next;
86     }
87 }
88

```


Main Class

```
1  package com.finalExam;
2
3  import java.util.Scanner;
4
5  public class MainProgram
6  {
7      static void menu()
8      {
9          System.out.println(x: "\n== Tax Payment Program ==");
10         System.out.println(x: "1. Pay Vehicle Tax");
11         System.out.println(x: "2. Display All Transaction & Total Transaction");
12         System.out.println(x: "0. Exit");
13         System.out.print(s: "Select: ");
14     }
15
16     Run | Debug | Run main | Debug main
17     public static void main(String[] args)
18     {
19         Scanner sc = new Scanner(System.in);
20         DLLVehicle dv = new DLLVehicle();
21         DLLTax dTax = new DLLTax();
22
23         dv.addVehicleData(new Vehicle(tnkb: "A1234AA", name: "Deras", type: "2 wheels", cc: 150, year: 2022, dueMonth: 3));
24         dv.addVehicleData(new Vehicle(tnkb: "B1234BB", name: "Wendry", type: "4 wheels", cc: 2000, year: 2021, dueMonth: 5));
25         dv.addVehicleData(new Vehicle(tnkb: "C1234CC", name: "Yoga", type: "2 wheels", cc: 90, year: 2023, dueMonth: 2));
26         dv.addVehicleData(new Vehicle(tnkb: "D1234DD", name: "Kuol", type: "4 wheels", cc: 3000, year: 2020, dueMonth: 8));
27
28         int select;
29         do
30         {
31             menu();
32             select = sc.nextInt();
33             sc.nextLine();
34         }
35     }
36 }
```

```

34     switch (select)
35     {
36         case 1:
37             System.out.print(s: "Input Vehicle TNKB: ");
38             String tnkb = sc.nextLine();
39
40             Vehicle sv = dv.searchVehicleTNKB(tnkb);
41
42             if (sv == null)
43                 System.out.println(x: "Vehicle is not found!!");
44             else
45             {
46                 System.out.print(s: "Input Month of Payment (1-12): ");
47                 int monthPay = sc.nextInt();
48                 sc.nextLine();
49
50                 Tax newTax = new Tax(sv, monthPay);
51                 dTax.addTaxTransaction(newTax);
52
53                 System.out.println(x: "Tax has been pay!!!");
54             }
55             break;
56
57         case 2:
58             dTax.displayAll();
59             break;
60
61         case 0:
62             System.out.println(x: "Thanks!!");
63             break;
64
65         default:
66             System.out.println(x: "Input Invalid!! Please, Try Again!");
67     }
68 }
69 while (select != 0);
70
71 sc.close();
72 }
73 }

```

Terminal Output/Debugging

```
== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 2
Tax Transaction is empty!!

== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 5
Input Invalid!! Please, Try Again!

== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 1
Input Vehicle TNKB: Z1234ZZ
Vehicle is not found!!
```

```
== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 1
Input Vehicle TNKB: A1234AA
Input Month of Payment (1-12): 5
Tax has been pay!!!
```

```
== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 1
Input Vehicle TNKB: B1234BB
Input Month of Payment (1-12): 3
Tax has been pay!!!
```

```
== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 1
Input Vehicle TNKB: C1234CC
Input Month of Payment (1-12): 7
Tax has been pay!!!
```

```
== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 1
Input Vehicle TNKB: D1234DD
Input Month of Payment (1-12): 8
Tax has been pay!!!
```

```
== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 2


| Code | TNKB    | Name   | Type     | CC   | Due Mon | Pay Mon | Nominal   | Fine     | Total Pay |
|------|---------|--------|----------|------|---------|---------|-----------|----------|-----------|
| 1    | A1234AA | Deras  | 2 wheels | 150  | 3       | 5       | Rp250000  | Rp50000  | Rp300000  |
| 4    | D1234DD | Kuol   | 4 wheels | 3000 | 8       | 8       | Rp1500000 | Rp0      | Rp1500000 |
| 2    | B1234BB | Wendry | 4 wheels | 2000 | 5       | 3       | Rp1000000 | Rp0      | Rp1000000 |
| 3    | C1234CC | Yoga   | 2 wheels | 90   | 2       | 7       | Rp100000  | Rp250000 | Rp350000  |


Total Transaction: Rp 3150000

== Tax Payment Program ==
1. Pay Vehicle Tax
2. Display All Transaction & Total Transaction
0. Exit
Select: 0
Thanks!!
```

Github

<https://github.com/guramedadar/Praktikum-Algoritma-Struktur-Data/tree/main/src/com/finalExam>